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(b) Equivalent mass of a oxidising agent $= \dfrac{\text{molar mass}}{\text{number of moles of electrons gained by one mole of the reducing agent}}$

$= \dfrac{292.2\ g\ mol^{-1}}{6\ eq\ mol^{-1}}$

$= \mathbf{48.7\ g\ eq^{-1}}$

5.

Element	**Percentage Composition**	**Atomic mass**	**Relative no. of atoms = Percentage / Atomic mass**	**Simple ratio**
C	54.55 %	12	54.55/12 = 4.55	4.55 / 2.27 = 2
H	9.09 %	1	9.09 / 1 = 9.09	9.09 / 2.27 = 4
O	36.36 %	16	36.36/16 = 2.27	2.27/2.27 = 1
Empirical formula (C_2H_4O)				

6.

Element	**Percentage Composition**	**Relative no. of atoms = Percentage / Atomic mass**	**Atomic mass= Percentage / Relative no. of atoms**	**Simple ratio**
X	32 %	2	16	4
Y	24 %	1	24	2
Z	44 %	0.5	88	1
Empirical formula ($X_4\ Y_2\ Z$)				

Calculated empirical formula mass $= (16\times4)+(24\times2)+88$

$= 64+48+88=200$

$n = \dfrac{\text{molar mass}}{\text{calculated empirical formula mass}}$

$\therefore n = \dfrac{400}{200}$

$= 2$

$\therefore$ **Molecular formula $(X_4Y_2Z)_2$** $=$ **$X_8Y_4Z_2$**